



Hypothesis Testing (continued)

Key terms

one-sample t -test, two-sample t -test, paired t -test, Chi-squared test for goodness of fit, Chi-squared test for independence

KSBs: K13, S11, S52, B1

7.1 More Statistical Tests

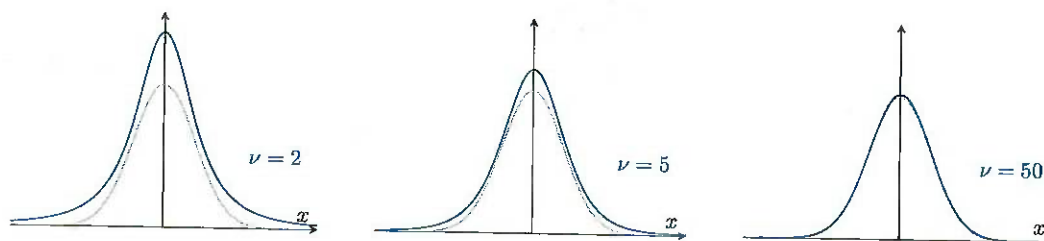
We have seen the z -test as an example of how hypothesis testing can be used to compare samples to a normal distribution. There are many other statistical tests that can be used instead, following the same process - formulating **hypotheses**, calculating a **test statistic** and obtaining a **probability** by comparing it to a known distribution.

7.2 t -tests

Recall that we previously used the Central Limit Theorem to study populations (with or without an underlying **normal distribution**), but that there were requirements that the sample size be sufficiently **large**. We also only really treated cases where we knew the **standard deviation** of the whole population. But it might be the case that we don't know the standard deviation, or that our sample sizes are too small - and in that case we can use the **t -distribution** and its associated **t -test**.

Definition 7.1 - t -distribution

The **Student's t -distribution** (or simply the t -distribution) t_ν is a continuous probability distribution that is **symmetric around zero** and **bell-shaped**. It is similar to the normal distribution, but depends on a parameter ν (the Greek letter nu), which is based on the sample size. The shape of the distribution depends on ν , and as ν gets larger it more closely approximates **$N(0, 1)$** .



The value of ν is the number of **degrees of freedom** for the test, which is one less than the sample size. For samples of size n , **$\nu = n - 1$** .

Note

The name 'Student' comes from the fact that the statistician who originally discovered the t -distribution, William Gossett, was working for the Guinness Brewery at the time of the discovery, and was not allowed to publish under his own name by request from his employer, so he was credited as 'Student'.

The t -distribution looks different from the normal distribution in the case of *small values of ν* and gives a more accurate approximation to the distribution of sample means for small sample sizes. We follow a very similar process to that for the *Z-test for normally distributed sample means*, except we use a different lookup table, in which we can find the statistic corresponding to a particular probability given a value of ν .

7.2.1 One-sample t -test

If we wish to compare a sample of data to a normal distribution with known *mean* but unknown *standard deviation*, we use a *one-sample t -test*.

Example 7.1.

The length of Hollywood films is known to be normally distributed. Let X be the length of a Hollywood film. In 2010, it was observed that the mean length of a film was *163.2* minutes long.

It is suspected that the length of films have been increasing. A sample of *7* films is taken and the mean of this sample is found to be *173.6* minutes with a standard deviation of *7.2* minutes.

We know the length of films is normally distributed, but the standard deviation is unknown, and the sample taken is *too small* to use a z -test. In this case, we can carry out a t -test.

$$H_0: \mu = 163.2$$

$$H_1: \mu > 163.2$$

→ Use a *one-tailed t -test* at the *5%* level with $\nu = 7 - 1 = 6$.

To calculate the test statistic, we use the same formula as for the z -test for the sampling distribution - except we can use the *standard deviation of the sample* rather than needing the population standard deviation.

$$t = \frac{\bar{x} - \mu}{\frac{S_x}{\sqrt{n}}} = \frac{173.6 - 163.2}{7.2/\sqrt{7}} = 3.82$$

Looking this value up in the table for the t -distribution, we go down to the row corresponding to $\nu = 6$ then look across and check the one-tailed row for probabilities. The value *3.82* lies between *3.707* and *5.208* meaning it corresponds to a probability between *0.005* and *0.001* which is below the significance level of 5% or 0.05.

Alternatively, we could check the table to see that for 5% significance on a 1-tail (upper tail) test, the critical value for the test statistic is *1.943* and the critical region is $t > 1.943$. Since $t = 3.82$ this is *in the critical region*.

This means we can *reject* the null hypothesis. There is *significant evidence* to suggest the length of a film has increased from 163.2 minutes.

t-test Probability Table

1-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
2-tail	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.025	0.010	0.002	0.001
1	0.000	1.000	1.376	1.963	3.078	6.314	12.710	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

A copy of this table will be included in the exam paper.

Exercise 7.1. A production line is designed to fill bottles with oil. The amount of oil placed in a bottle is normally distributed and the mean is set to 100 ml. The amount of oil, x ml, in each of 8 randomly selected bottles is recorded, and the following statistics are obtained.

$$\bar{x} = 92.875, \sigma_x = 8.3055$$

The supervisor believes that the mean amount of oil placed in a bottle is less than 100 ml. Stating your hypotheses clearly, test, at the 5% significance level, whether or not their belief is supported.

7.2.2 Two-sample t -test

There is also a *two-sample* version of the t -test, which allows us to compare two samples and check if they came from underlying distributions *with the same mean*. This is a commonly used version of the t -test, and has many useful applications.

The test statistic is more complicated to calculate, as it involves combining the parameters from *both samples*, - but it can be computed using software (see the end of the next section).

7.2.3 Paired t -test

We can also use a t -test to compare *paired data* - for the case where you have two samples in which each entry is *paired with* a corresponding entry in the other sample. For example, if a group has taken two different exams and you would like to compare their performance, you would have paired data from each set of exam results, and could use a *paired t -test* to compare the *means*.

Example 7.2. In a trial for new cough medicine, a random sample of 8 healthy patients were given steadily increasing doses of a pepper extract until they started coughing. The level of pepper *that triggered the coughing* was recorded. Each patient completed the trial after taking a *standard cough medicine* and, at a later time, after taking the *new medicine*. The results are given in the table below.

Patient	A	B	C	D	E	F	G	H
Standard Medicine	46	12	18	31	23	16	27	9
New Medicine	53	16	13	49	11	34	38	22

Assuming that the differences in pepper levels between the two groups are normally distributed, carry out a test at the 5% level to determine which medicine you would recommend (if any).

Let X be the level of pepper that triggers coughing in a patient taking the standard medicine.

Let Y be the level of pepper that triggers coughing in a patient taking the new medicine.

Let $D = X - Y$ which we assume is normally distributed.

$$H_0: \mu_D = 0; H_1: \mu_D \neq 0$$

Use a two-tailed paired t -test at the 5% level using $v = 8 - 1 = 7$.

The differences between each pair of values are $-7, -4, 5, -18, 12, -18, -11$ and -13 . We can calculate the mean $\bar{d} = -6.75$ and standard deviation as $s_d = 10.74$. So the test statistic is:

$$t = \frac{\bar{d} - \mu}{s_d / \sqrt{n}} = \frac{-6.75 - 0}{10.74 / \sqrt{8}} = -1.78.$$

The critical value for a two-tailed test with $v = 7$ is 2.365 so the critical region will be values $t < -2.365$, and $t > 2.365$. The test statistic is not in the critical region which means the result is not significant. We do not reject H_0 as there is insufficient evidence to suggest that there is a difference in the mean level of pepper which triggers coughing between the standard and the new medicine.

Exercise 7.2.

- (a) Two methods of extracting juice from an orange are to be compared. Eight oranges are halved. One half of each orange is chosen at random and allocated to Method A and the other half is allocated to Method B. The amounts of juice extracted, in ml, are given in the table.

Orange	1	2	3	4	5	6	7	8
Method A	29	30	26	25	26	22	23	28
Method B	27	25	28	24	23	26	22	25

One statistician suggests performing a two-sample t -test to investigate whether or not there is a difference between the mean amounts of juice extracted by the two methods. Another statistician suggests analysing these data using a paired t -test.

- Explain why the paired t -test is a more appropriate test to use.
- Carry out this test.

- (b) As part of an investigation into the effectiveness of solar heating, a pair of houses was identified which had the same mean weekly fuel consumption. One of the houses was then fitted with solar heating and the other was not. Following the fitting of the solar heating, a random sample of 9 weeks was taken and the table below shows the weekly fuel consumption for each house.

Week	1	2	3	4	5	6	7	8	9
Without solar heating	19	19	18	14	6	7	5	31	43
With solar heating	13	22	11	16	14	1	0	20	38

- (i) Test whether or not there is evidence that the solar heating reduces the mean weekly fuel consumption.
 (ii) State an assumption about weekly fuel consumption that is required to carry out this test.

For each type of t -test there is a formula for the t -statistic which can then be looked up in the table. In general, we can use software find the *probability* and compare it to the significance level for the test we are performing.

Function: T.TEST

The function =T.TEST() returns the probability for a t -test. It takes four inputs: two ranges for the two datasets, a value for the number of tails (1 or 2), and a fourth variable for the type of t -test, for which:

- a value of 1 corresponds to a paired t -test
- a value of 2 corresponds to a two-sample t -test when the variance of the two samples are equal
- a value of 3 corresponds to a two-sample t -test when the variance of the two samples are different.

7.3 Chi-Squared Test

The Chi-squared test is a type of statistical test used for *categorical data* - when your observations are grouped into categories, and the data you're working with is the *number of observations* that fall into each category. You might wish to study the distribution across the different categories, by comparing it to a *known distribution*, or a *different set of data* grouped into the same categories.

The test statistic for this test is denoted by χ^2 - this is the Greek letter χ (*chi*) which is pronounced 'kye'.

The test, much like the other tests we've seen, involves calculating a *test statistic* and using a lookup table, or inputting the values into a function, to get a *probability* which we can compare to a *p-value*.

There are two main types of Chi-squared test: a test that looks at how well a *distribution fits* with a pre-specified distribution, and for a test for *independence* between two different groupings of two variables.

In both tests, we compare the *values in the dataset* - called the *observed values* - with *those we would expect to see* given the distribution we're looking for - called the *expected values* - and use the difference between these to get a test statistic.

Definition 7.2 - χ^2 Test Statistic

The test statistic for the Chi-squared test is calculated as:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where ν is the number of *degrees of freedom* for the test, O_i are the *observed values* and E_i are the *expected values*.

Degrees of freedom are calculated slightly differently for different types of test.

7.3.1 Chi-squared Test for goodness of fit

We can use a version of the Chi-squared test to compare the distribution of categorical data to a known given distribution. It might be that a sample has been taken from *a larger population* and we wish to see how well the distribution across categories in the sample fits with the whole population; or, it may be we have been told there is a distribution we are aiming for, and want to check how well the data *fits with it*.

Definition 7.3 - Degrees of Freedom (χ^2 Test for Goodness of Fit)

The number of degrees of freedom in a χ^2 test for goodness of fit is *the number of categories in the data minus 1*.

Note

This is because if we know the total number of observations, and had all but one of the values for the categories, we could work out the missing value. The final category does not have 'freedom', since its value is determined by the values of the others.

Chi-Squared Test Probability Table

DF	1.00	0.99	0.98	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.03	0.01	0.01
1	0.0000	0.0002	0.0010	0.0039	0.0158	0.1015	0.4549	1.3233	2.7055	3.8415	5.0239	6.6349	7.8794
2	0.0100	0.0201	0.0506	0.1026	0.2107	0.5754	1.3863	2.7726	4.6052	5.9915	7.3778	9.2103	10.597
3	0.0717	0.1148	0.2158	0.3519	0.5844	1.2125	2.3660	4.1083	6.2514	7.8147	9.3484	11.345	12.838
4	0.2070	0.2971	0.4844	0.7107	1.0636	1.9226	3.3567	5.3853	7.7794	9.4877	11.143	13.277	14.860
5	0.4117	0.5543	0.8312	1.1455	1.6103	2.6746	4.3515	6.6257	9.2364	11.071	12.833	15.086	16.750
6	0.6757	0.8721	1.2373	1.6354	2.2041	3.4546	5.3481	7.8408	10.645	12.592	14.449	16.812	18.548
7	0.9893	1.2390	1.6899	2.1674	2.8331	4.2549	6.3458	9.0372	12.017	14.067	16.013	18.475	20.278
8	1.3444	1.6465	2.1797	2.7326	3.4895	5.0706	7.3441	10.219	13.362	15.507	17.535	20.090	21.955
9	1.7349	2.0879	2.7004	3.3251	4.1682	5.8988	8.3428	11.389	14.684	16.919	19.023	21.666	23.589
10	2.1559	2.5582	3.2470	3.9403	4.8652	6.7372	9.3418	12.549	15.987	18.307	20.483	23.209	25.188
11	2.6032	3.0535	3.8158	4.5748	5.5778	7.5841	10.341	13.701	17.275	19.675	21.920	24.725	26.757
12	3.0738	3.5706	4.4038	5.2260	6.3038	8.4384	11.340	14.845	18.549	21.026	23.337	26.217	28.300
13	3.5650	4.1069	5.0088	5.8919	7.0415	9.2991	12.340	15.984	19.812	22.362	24.736	27.688	29.819
14	4.0747	4.6604	5.6287	6.5706	7.7895	10.165	13.339	17.117	21.064	23.685	26.119	29.141	31.319
15	4.6009	5.2294	6.2621	7.2609	8.5468	11.037	14.339	18.245	22.307	24.996	27.488	30.578	32.801
16	5.1422	5.8122	6.9077	7.9617	9.3122	11.912	15.339	19.369	23.542	26.296	28.845	32.000	34.267
17	5.6972	6.4078	7.5642	8.6718	10.085	12.792	16.338	20.489	24.769	27.587	30.191	33.409	35.718
18	6.2648	7.0149	8.2308	9.3905	10.865	13.675	17.338	21.605	25.989	28.869	31.526	34.805	37.156
19	6.8440	7.6327	8.9065	10.117	11.651	14.562	18.338	22.718	27.204	30.144	32.852	36.191	38.582
20	7.4338	8.2604	9.5908	10.851	12.443	15.452	19.337	23.828	28.412	31.410	34.170	37.566	39.997
21	8.0337	8.8972	10.283	11.591	13.240	16.344	20.337	24.935	29.615	32.671	35.479	38.932	41.401
22	8.6427	9.5425	10.982	12.338	14.041	17.240	21.337	26.039	30.813	33.924	36.781	40.289	42.796
23	9.2604	10.196	11.689	13.091	14.848	18.137	22.337	27.141	32.007	35.172	38.076	41.638	44.181
24	9.8862	10.856	12.401	13.848	15.659	19.037	23.337	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	19.939	24.337	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	20.843	25.336	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	21.749	26.336	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	22.657	27.336	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	23.567	28.336	33.711	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	24.478	29.336	34.800	40.256	43.773	46.979	50.892	53.672

A copy of this table will be included in the exam paper.

Example 7.3.

A casino die is rolled **1200 times** and the outcome of each roll is recorded. The casino owner would like to know if this is a **fair die**, since it is being used for high-stakes gameplay.

H_0 : the die is fair, so we will see a uniform distribution

H_1 : the die is biased towards certain numbers more than others.

The **observed numbers** of each dice roll are given in the table below, and the expected values can be calculated by multiplying the **probability of each result** (in this case, uniformly $\frac{1}{6}$) by the **total number of rolls (1200)**.

Roll						
Obs. count	179	211	181	186	224	219
Exp. frequency	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
Exp. count	200	200	200	200	200	200
$\frac{(O-E)^2}{E}$	$\frac{-21^2}{200} = 2.205$	$\frac{11^2}{200} = 0.605$	$\frac{-19^2}{200} = 1.805$	$\frac{-14^2}{200} = 0.98$	$\frac{24^2}{200} = 2.88$	$\frac{19^2}{200} = 1.805$

To calculate the Chi-squared statistic, we need to complete the bottom row, by calculating $\frac{(O-E)^2}{E}$ for each category. The statistic is then given by the sum of the entries in the last row, namely

$$2.205 + 0.605 + 1.805 + 0.98 + 2.88 + 1.805 = 10.28$$

The number of degrees of freedom is the number of faces on the die, minus one - so $\sqrt{6-1} = 5$.

Using the reference table for the Chi-squared test, we see that 10.28 corresponds to a p-value between 0.05 and 0.10. Since this is **greater than 5%** we **do not reject** the null hypothesis that the die is fair - while we haven't seen exactly equal numbers of each value, this is within the margins we have specified. The distribution we saw is not that unexpected for a fair die (it would be observed **more than 5% of the time**).

Example 7.4.

A survey is conducted across a large company, and the HR department collects **1000 responses**. The survey relates to diversity, and HR are keen to make sure that the sample of employees surveyed reflects the ethnic background of employees in the company generally.

H_0 : the distribution of ethnicities in the survey is reflective of the employee pool as a whole, and is a true random sample

H_1 : the distribution is very different from the employee pool as a whole.

The number of staff members from each category who responded to the survey are given in the table below, and the proportion of these ethnicities in the overall staff are also given. The expected values can be calculated by multiplying the **proportion of staff** in each category by the **total number of staff surveyed (1000)**.

Background	Asian	Black	Mixed	White	Other
Obs. count	90	25	27	844	14
Exp. frequency	10%	4%	3%	81%	2%
Exp. count	100	40	30	810	20
$\frac{(O-E)^2}{E}$	$\frac{-10^2}{100} = 1$	$\frac{-15^2}{40} = 5.625$	$\frac{-3^2}{30} = 0.3$	$\frac{34^2}{810} = 1.427$	$\frac{(-6)^2}{20} = 1.8$

To calculate the Chi-squared statistic, we need to complete the bottom row, by calculating $\frac{(O-E)^2}{E}$ for each category. The statistic is then given by the sum of the entries in the last row, namely

$$\chi^2 = 1 + 5.625 + 0.3 + 1.427 + 1.8 = 10.152.$$

The number of degrees of freedom is the number of categories, minus one - so $\nu = 5 - 1 = 4$.

Using the reference table for the Chi-squared test, we see that 10.152 corresponds to a p-value between 0.03 and 0.05. Since this is less than 0.05 we reject the null hypothesis that the sample is representative of the employee population - a true random sample would have a distribution like this less than 5% of the time.

Exercise 7.3. (a) A discrete random variable X has the following probability distribution.

X	8	15	16	28	31
$P(X = x)$	0.4861	0.1344	0.1245	0.1158	0.1392
Exp. count.					

Given that a total of 300 observations were made, find the expected count for each value.

(b) This random variable is used to model the following data:

x	8	15	16	28	31
Obs. count	23	28	24	29	21
$\frac{(O-E)^2}{E}$					

- Find the value of χ^2 and state the value of ν .
- Is the data a good fit?

7.3.2 Chi-squared Test for Independence

If we have *two variables* in our categorical data, we may wish to compare the proportions of observations that fall into each category and see if there is any *association between them*. Provided the entries in the table represent *randomly sampled* observations from a population, we can use a χ^2 *test for independence* to investigate the significance of any differences.

Once calculated, the test statistic uses the same lookup table as the χ^2 test for goodness of fit.

Definition 7.4 - Degrees of Freedom (χ^2 Test for Independence)

The number of degrees of freedom ν in a χ^2 test for independence is the number of categories in one variable, minus one, multiplied by the number of categories in the other variable, minus one. For example, if we have *5* categories for one variable and *3* for the other, ν would be given by $(5-1)(3-1) = 4 \times 2 = 8$.

To calculate the expected values for this test, we can use the *proportions of the data falling in each category* to work out how many we would expect to see in each combination of categories.

Example 7.5. Recall our data about people visiting a mountain lodge, which we presented in a contingency table.

	Hiker	Non-hiker	Total
Skiier	178	45	223
Non-skiier	132	18	150
Total	310	63	373

From the table, we can see that *223 of the 373* people were there for skiing, and *150 of the 373* were not there for skiing. If these variables were independent, we would expect this ratio to also be present if we looked only at the hikers present - that is, we'd expect *$\frac{223}{373}$ of the hikers* to also be skiiers.

To calculate the number of skiiers this represents, we can calculate

$$\frac{223}{373} \times 310 = 185.3 \sim 185$$

For each entry in the table (not including the totals) we can calculate the expected values in this way, and obtain:

	Hiker (E)	Non-hiker (E)
Skiier	$\frac{223}{373} \times 310 = 185$	$\frac{223}{373} \times 63 = 39$
Non-skiier	$\frac{150}{373} \times 310 = 125$	$\frac{150}{373} \times 63 = 26$

Comparing these values to the original table, they look to be *broadly similar* to the actual values observed.

Note

To perform this test, we need at least *five observations* for the expected values, in each category - if this is not the case, it may be necessary to *combine categories together*. This reduces the value of ν , but if otherwise the test *cannot be performed at all*, this is preferable.

To quantify exactly how similar the expected frequencies are to the observed frequencies, we can use the χ^2 distribution.

Example 7.6. 600 people visiting a cinema are surveyed to find out what kind of film they were there to see, and whether or not they bought snacks. We will perform a χ^2 test for independence, at the 5% significance level.

H_0 : the choice of film type and snacks is independent
 H_1 : there is a relationship between them.

Movie type	Snacks	No snacks	Total
Action	50	75	125
Comedy	125	175	300
Family	90	30	120
Horror	45	10	55
Total	310	290	600

For each combination of movie type and snack/no snack, we can calculate the expected values, using the totals for those rows/columns.

Movie type	Snacks (O)	Snacks (E)	No snacks (O)	No snacks (E)
Action	50	$\frac{125}{600} \times 310 = 64.58$	75	$\frac{125}{600} \times 290 = 60.42$
Comedy	125	$\frac{300}{600} \times 310 = 155$	175	$\frac{300}{600} \times 290 = 145$
Family	90	$\frac{120}{600} \times 310 = 62$	30	$\frac{120}{600} \times 290 = 58$
Horror	45	$\frac{55}{600} \times 310 = 28.42$	10	$\frac{55}{600} \times 290 = 26.58$

Recall that the χ^2 test statistic is given by $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$. We can calculate the value of $\frac{(O-E)^2}{E}$ for each cell in the table:

Movie type	Snacks (O)	Snacks (E)	$\frac{(O-E)^2}{E}$	No snacks (O)	No snacks (E)	$\frac{(O-E)^2}{E}$
Action	50	$\frac{125}{600} \times 310 = 64.58$	$\frac{(50-64.58)^2}{64.58} = 3.29$	75	$\frac{125}{600} \times 290 = 60.42$	$\frac{(75-60.42)^2}{60.42} = 3.52$
Comedy	125	$\frac{300}{600} \times 310 = 155$	$\frac{(125-155)^2}{155} = 5.81$	175	$\frac{300}{600} \times 290 = 145$	$\frac{(175-145)^2}{145} = 6.21$
Family	90	$\frac{120}{600} \times 310 = 62$	$\frac{(90-62)^2}{62} = 12.65$	30	$\frac{120}{600} \times 290 = 58$	$\frac{(30-58)^2}{58} = 13.52$
Horror	45	$\frac{55}{600} \times 310 = 28.42$	$\frac{(45-28.42)^2}{28.42} = 9.68$	10	$\frac{55}{600} \times 290 = 26.58$	$\frac{(10-26.58)^2}{26.58} = 10.35$

To find the overall test statistic, we add these values for each cell:

$$3.29 + 3.52 + \dots + 10.35 = 65.03$$

The degrees of freedom are given by $\nu = (4-1)(2-1) = 3 \times 1 = 3$. Looking up in the table, for $\nu = 3$ the critical value for 5% significance is 7.8147. Since $65.03 > 7.8147$ we can reject the null hypothesis and conclude that there is a relationship between movie type and snack purchases.

Example 7.7. The operations manager of a company that manufactures tires wants to determine whether there are any differences in the quality of work among the three daily shifts. She randomly selects 496 tires and carefully inspects them. Each tire is either classified as perfect, satisfactory, or defective, and the shift that produced it is also recorded.

We will perform a χ^2 test for independence, at the 5% significance level.

H_0 : the shift and quality are independent

H_1 : there is a relationship between the shift and tyre quality.

	Perfect	Satisfactory	Defective	Total
Shift 1	106	124	1	231
Shift 2	67	85	1	153
Shift 3	37	72	3	112
Total	210	281	5	496

For each cell we can calculate the expected value, and the value of $\frac{(O-E)^2}{E}$.

	Perf. (O)	Perf. (E)	$\frac{(O-E)^2}{E}$	Satisf. (O)	Satisf. (E)	$\frac{(O-E)^2}{E}$	Defect. (O)	Defect. (E)	$\frac{(O-E)^2}{E}$
Shift 1	106	97.8	0.69	124	130.87	0.36	1	2.33	0.76
Shift 2	67	64.78	0.08	85	86.68	0.03	1	1.54	0.19
Shift 3	37	47.42	2.29	72	63.45	1.15	3	1.13	3.1

The degrees of freedom are $\nu = (3-1)(3-1) = 2 \times 2 = 4$.
 $\frac{(O-E)^2}{E}$ values:

$$0.69 + 0.08 + \dots + 3.1 = 8.65$$

The χ^2 statistic is the sum of the

Looking up in the table, for $\nu = 4$ the critical value for 5% significance is 9.4877. Since $8.65 < 9.4877$ we can accept the null hypothesis and conclude that there is no relationship between the shift and the quality of tyres produced.

Note

Since this example has a *smaller expected value than 5* in multiple categories, the outcome of the test would not be considered reliable. In practice, you could combine the 'Satisfactory' and 'Defective' columns (your categories could become 'Perfect' and 'Imperfect') by adding the total and perform the analysis on this data, assuming this would still give useful information. An alternative would be to collect a *larger sample of data*.

Function: CHISQ.TEST (E/S)

The =CHISQ.TEST() function will perform the Chi-squared test and return the p -value. It takes as inputs two ranges - the observed and the expected values - and doesn't need to be told ν as it calculates this from the size of the ranges.

It can be used for either type of test, as long as you can give it the values of O_i and E_i for each category. For tests for independence, the ranges can have multiple rows/columns.

Exercise 7.4.

(a) For the data in Example 7.5, perform a χ^2 test for independence, at 5% significance.

	Hiker (O)	Hiker (E)	$\frac{(O-E)^2}{E}$	Non-hiker (O)	Non-hiker (E)	$\frac{(O-E)^2}{E}$
<i>Skiier</i>	178	185		45	39	
<i>Non-skiier</i>	132	125		18	26	

- (b) The following table shows a random sample of 200 customers in a café which only serves goths and emos, and how many ordered tea, coffee or chocolate.

	Tea	Coffee	Chocolate
Goth	57	26	11
Emo	42	47	17

Use a χ^2 test, at the 5% level of significance, to investigate whether there is any association between the type of drink ordered and music preference.

	Tea			Coffee			Chocolate		
Goth	57			26			11		
Emo	42			47			17		

7.4 Choosing the right test

For each of the tests we have seen so far, the table below outlines the conditions required for each test, and circumstances in which they might be used. You will encounter many other types of statistical test which can be used for hypothesis testing, and depending on the circumstances you might find more suitable tests not on this list.

Test	Description	Requirements	Use case
z-test of a single observation x	calculate $z = \frac{x-\mu}{\sigma}$, using population μ and σ ; look up in ND table and compare to p -value	normal distribution Known population μ, σ	checking if a single value is from a given normal distribution
One-sample z-test	calculate z-score for given sample, using $x = \bar{x}$, population μ and standard error $\frac{\sigma}{\sqrt{n}}$; look up in ND table and compare to p -value	Known population μ, σ random sample $n > 30$ OR normal dist.	to check if a (large enough) sample has the same mean as the underlying population
One-sample t -test	calculate t -score for given sample, using $x = \bar{x}$, population μ and standard error $\frac{\sigma}{\sqrt{n}}$; look up in t -table and compare to p -value	Known population μ random sample.	to check if a sample has the same mean as the underlying population, when samples are too small for a z-test or population σ is unknown
Two-sample t -test	calculate t -score from two given sample means; find corresponding probability and compare to p -value	Known pop. μ random samples	to check if two samples came from underlying distributions with the same mean
Paired t -test	find the difference between the two variables, D ; calculate μ_D and s_D ;	paired data random sample D normally dist.	to compare two sets of observations about the same sample
χ^2 test for goodness of fit	calculate expected counts for each category; calculate $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ and look up in row of χ^2 table corresponding to ν	Categorical data random sample Known distribution	to compare a dataset to a known distribution
χ^2 test for independence	calculate expected counts for each category; calculate $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ and look up in row of χ^2 table corresponding to ν	Categorical data (two variables) random samples $E_i > 5$ for all i .	to test whether two variables are independent